Two-Rate Energy-Authorized Residual MPC for Impedance-Causal Physical Interaction

Abstract

Residual model predictive control (MPC) can improve disturbance rejection around a compliant robot, but its corrective wrench is an active interaction port. A passivity check performed only when the MPC updates does not authorize the wrench held between updates, and strict time-domain passivity control can remove most of the predictive correction. We study a rate-separated architecture in which a motion-to-wrench impedance remains the physical nominal controller, a 100 Hz MPC proposes only a translational residual wrench, and a 1 kHz projection jointly enforces a residual-energy budget and the complete joint-torque envelope. This is not presented as the first combination of impedance, MPC, and passivity: passive model-predictive impedance, predictive variable-impedance/admittance, and energy-tank controllers are established. The investigated contribution is the explicit residual-port authorization and its inter-update realization. A continuous-time composition proposition and a fast-sample energy-floor lemma state the certificate boundary. An equation-faithful translational generalization of the Hannaford–Ryu time-domain passivity observer/controller provides an external baseline. In a torque-controlled 7-DoF Franka FR3 MuJoCo benchmark, 20 matched trials randomize passive-wall properties, disturbance amplitude, and phase. The proposed method has zero tank-floor violations, zero nominal-torque infeasibilities, and zero QP failures. It reduces residual-position RMS by 23.7% relative to passive impedance and by 35.4% relative to strict time-domain passivity control, while remaining 19.5% worse than energetically unguarded MPC. A force-separation study shows that 50% leakage of intentional force into the disturbance estimate increases intentional-axis fidelity error by 28.2%; adding estimator delay, colored noise, and a constant velocity-estimate bias leaves the safety certificates untouched but still degrades task fidelity, most from the velocity bias. A manager-rate sweep (20/50/100 Hz, fixed 0.20 s lookahead) finds no universal best rate: the slower manager reduces correction chatter and wins under oscillatory wall-contact disturbance, while the faster manager wins at tracking a sustained intentional push; safety invariants hold at every rate tested. An anticipatory variant that forecasts a known-frequency disturbance through the horizon, rather than holding it constant, is tested against the frozen hold: a controlled ablation finds it statistically neutral when the disturbance is genuinely harmonic, but measurably worse (1.6-1.75%) once any non-harmonic content − a brief pulse, or contact − enters the estimate, a gap a longer horizon widens rather than closes, at added modeling and engineering complexity for no compute saving. The results support the two-rate authorization mechanism in full rigid-body simulation, but not hardware-level safety or universal superiority.

Keywords: impedance control, residual model predictive control, passivity, energy tank, multi-rate control, physical human–robot interaction

1. Introduction

Impedance control exposes a desired motion-to-wrench relation at the robot's physical interaction port [1]. Predictive control offers complementary benefits: disturbance preview, finite-horizon allocation, and explicit actuator constraints. The difficulty is energetic. An additive predictive wrench can inject energy even when the nominal impedance is passive, and a decision made at the slow optimizer rate need not remain admissible as velocity and actuator headroom evolve during the held interval.

The architecture contains three components:

1. a 1 kHz Cartesian impedance that produces the complete nominal joint torque;
2. a 100 Hz MPC that proposes an additive translational residual wrench; and
3. a 1 kHz realization layer that scales the held proposal against measured velocity, tank energy, and complete joint-torque headroom.

Section 2 walks through this loop once in plain language before any equations. Section 4 then builds the physical model the loop runs on: robot dynamics, the impedance law that is the actual physical nominal, the admittance law used only as an analytical reference, and the error dynamics the MPC predicts. Section 5 gives the MPC itself and the two ways its proposal is authorized before reaching the actuator.

The contributions are:

- a causal and energetic decomposition separating nominal impedance, intentional-response reference, and residual-wrench authorization;
- a continuous-time composition result and a discrete fast-sample invariant that include the realized, not merely requested, residual wrench;
- a torque-controlled 7-DoF FR3 benchmark with full mass matrix, Jacobian, gravity/Coriolis bias, orientation hold, null-space control, and per-joint torque limits;
- an equation-faithful Hannaford–Ryu passivity-observer/controller (PO/PC) baseline sharing the same nominal controller and raw MPC proposal; and
- a direct force-separation leakage study that measures degradation of the intentional interaction response.

The paper remains a simulation study. Its claim is a reusable realization interface and evidence for inter-update authorization, not a new passivity or MPC principle.

2. How the Two-Rate Loop Works

This section describes the loop once, in words, before any of it is formalized in Section 4 and Section 5.

Fast loop, at the robot's own servo rate, 1 kHz. The physical nominal controller is an impedance law: it looks at the current position and velocity error and computes a wrench directly, the way a stiff spring-damper would. This wrench is converted to joint torque and, added to it, the *residual* wrench the slow loop most recently published (converted to torque at the *current* configuration, not cached from when it was computed). The result is scaled if needed so it never asks for more torque or more stored energy than is available, and only then sent to the robot. This last step is the fast authorization; it runs every 1 kHz tick regardless of what the slow loop is doing.

Slow loop, every manager period, 100 Hz, much slower than the fast loop but still ten times a second. The manager does not touch the impedance law. Instead it asks: if the impedance controller keeps doing what it's doing, and a human keeps pushing the way they currently appear to be pushing, will the resulting torque and stored energy stay legal over the next few fast-loop ticks? It rolls the impedance law's own behavior forward, and if that rollout would leave the actuator or the energy budget, it solves a small optimization for the smallest additive wrench correction that keeps the rollout legal. If nothing is about to go wrong, the correction is exactly zero and the impedance law runs unmodified.

Why two rates, and why authorize twice. The manager is slow because predicting several steps ahead and solving an optimization problem costs computation; running it at the full 1 kHz servo rate would be wasteful when the physical dynamics only drift meaningfully over tens of milliseconds. But a wrench correction computed once and held for several fast-loop ticks is a promise made about the future — by the time it is actually applied, the measured velocity or the remaining energy may have moved. The fast authorization is what keeps that promise honest between manager updates: it recomputes the torque conversion, the energy ledger, and the actuator margin every single tick, and scales the held correction down (never up) if reality has moved since the manager last checked. Sections 4 and 5 give the exact equations behind each of these three pieces; Section 6 shows what changes when the fast authorization is switched off.

3. Relation to Existing Work

The broad combination of impedance, MPC, and passivity is established. Cao, Cheng, and Li use a bottom variable-impedance controller and a top MPC that computes complementary torque under a stored-energy constraint; they prove passivity and feasibility and validate on a Franka Panda [2]. This is the closest architectural precedent and rules out novelty claims based only on stacking an impedance loop and predictive correction.

Predictive impedance adaptation is also mature. Haninger, Hegeler, and Peternel optimize trajectory and impedance using learned interaction models [3]. Xue *et al.* combine predictive variable impedance, environment estimation, robustness, and passive switching [4]. Shen *et al.* embed a passivity index in a predictive variable-impedance controller for a hydraulic manipulator [5]. Mahfouz *et al.* optimize admittance parameters with passivity constraints and validate with seven participants [6]. These methods optimize or schedule the rendered compliance; the present method fixes the physical nominal impedance and authorizes a separate residual wrench.

Energy supervision predates all of these predictive controllers. Hannaford and Ryu's time-domain PO/PC measures sampled port energy and adds exactly the damping required to eliminate generated energy [7]. Ferraguti, Secchi, and Fantuzzi use energy tanks to render time-varying stiffness passively [8], and related layered energy-tank architectures have been demonstrated in robotic surgery [9]. Guo *et al.* introduce ultimate passivity, switching between performance and conservative modes while retaining an ultimate energy bound [10].

These closest architectures differ along the same three axes: what quantity is optimized or supervised, what mechanism enforces energy safety, and how the method was validated. Hannaford and Ryu's PO/PC [7] supervises an arbitrary sampled port with zero-reference damping, validated on haptic hardware. Tank-based impedance methods [8], [9] instead supervise time-varying interaction behavior through stored energy, validated on robot prototypes. Passive model-predictive impedance [2] optimizes complementary torque under a predicted stored-energy constraint, validated on a Franka experiment. Predictive impedance/variable-impedance methods [3]–[5] optimize the trajectory and/or impedance itself under task-specific constraints or passivity, validated on robot experiments. Predictive admittance [6] optimizes admittance parameters under an embedded passivity constraint, validated with a Jaco-2 and seven participants. Ultimate passivity [10] instead switches controller mode to retain an ultimate energy bound, validated on impedance/admittance robots. This study optimizes an additive residual wrench through a 100 Hz proposal and a 1 kHz energy/torque projection, validated in 7-DoF rigid-body simulation.

The remaining question is therefore narrow: can a predictive residual wrench be given a finite, replenishable energy budget and realized at a faster rate without altering the nominal impedance or violating the complete torque interface?

4. Robot Dynamics and Control Architecture

This section builds the physical model underneath Section 2's informal walkthrough, in the order the loop actually needs it: the robot's own dynamics first (4.A), then the impedance law that is the real physical controller (4.B), then the admittance law that is *not* used physically but supplies the analytical reference the residual tracks (4.C), and finally the error-coordinate model the MPC of Section 5 actually predicts (4.D). The notation below collects every symbol introduced along the way.

Notation.

- $q, \dot{q}$ – joint position, velocity (rad, rad/s)
- $M(q), h(q, \dot{q})$ – joint mass matrix, gravity/Coriolis bias (kg · m², N · m)
- $J_v(q), J_\omega(q)$ – translational, rotational task Jacobian
- τ – commanded joint torque (N · m)
- $F = F_h + d + F_e$ – total task-space force: intentional, rejectable, environment (N)
- Λ – translational operational (task-space) inertia (kg)
- $p, v = J_v\dot{q}$ – task position, velocity (m, m/s)
- $e = p - p_0$ – position error against the fixed nominal pose p_0 (m)
- K_I, D_I – rendered impedance stiffness, damping (N/m, N · s/m)
- F_I – impedance-law wrench, the physical nominal (N)
- τ_I – complete nominal joint torque: impedance + orientation + posture (N · m)
- x_I – admittance-law reference trajectory, analytical only, never commanded (m)
- $z = e - x_I$ – residual: how far the real error is from the admittance reference (m)
- F_r – MPC's proposed residual wrench (N)
- H – impedance storage, an energy-like Lyapunov function (J)
- E – residual-energy tank ledger (J)

4.A Joint and task dynamics

For the 7-DoF arm,

$$M(q)\ddot{q} + h(q,\dot{q}) = \tau + J_v(q)^\top F, \qquad F = F_h + d + F_e, \tag{1}$$

in words: joint torque and task-space force together produce joint acceleration through the usual rigid-body dynamics; h is the complete gravity/Coriolis bias, F_h is intentional human force, d is a rejectable force component, and F_e is the passive environment wrench. The translational operational inertia is

$$\Lambda = (J_v M^{-1} J_v^\top)^{-1}, \tag{2}$$

in words: Λ is how much task-space mass the robot presents at the end-effector once the whole arm's inertia is projected through the current Jacobian — it is what converts a task-space force into a task-space acceleration, and it changes with configuration. The MuJoCo implementation evaluates M, h, J_v, and the rotational Jacobian J_ω from the current nonlinear state at every 1 kHz sample.

4.B Impedance control: the physical nominal

Impedance control commands a wrench directly from the measured motion error — it is *error-to-force* causal. For fixed nominal position p_0 and orientation R_0, define $e = p - p_0$, $v = J_v\dot{q}$, and

$$F_I = -K_I e - D_I v. \tag{3}$$

in words: the physical nominal controller behaves like a spring-damper bolted between the end-effector and the fixed nominal pose — the further away or the faster it moves, the harder it pulls back. This is the actual physical controller running at 1 kHz; nothing else in this paper commands the robot directly. The complete nominal torque is

$$\tau_I = h + J_v^\top F_I + J_\omega^\top F_R + N_v^\top \tau_0, \tag{4}$$

where F_R holds orientation and the dynamically consistent translational null-space projector N_v contains posture regulation. The residual wrench proposed by the MPC (Section 5) is added through the same physical channel, never a separate one:

$$\tau = \tau_I + J_v^\top F_r. \tag{5}$$

Ignoring the disclosed auxiliary-task leakage, the translational storage

$$H = \tfrac{1}{2} v^\top \Lambda v + \tfrac{1}{2} e^\top K_I e \tag{6}$$

in words: H is the impedance law's own energy-like quantity — kinetic energy in the task-space inertia plus potential energy in the virtual spring — and it has the familiar local power form

$$\dot{H} \le F^\top v - v^\top D_I v + F_r^\top v. \tag{7}$$

in words: storage grows from external work $F^\top v$, shrinks from the impedance law's own damping $-v^\top D_I v$, and the last term, $F_r^\top v$, is the power the residual wrench injects or removes — this is the port that Section 5.2's tank must authorize, because nothing about the impedance law itself limits it. Equation (7) is used for controller construction; the paper does not elevate the varying-Λ, regularized, sampled implementation to an exact global storage identity.

4.C Admittance control and the analytical intentional reference

A different, and in this literature very common, way to structure a compliant controller is *admittance* control: instead of mapping error to force, integrate a force-to-motion law forward to get a reference trajectory, then have an inner loop track that trajectory. Concretely,

$$M_I \ddot{x}_I + D_I \dot{x}_I + K_I x_I = F_h, \tag{8}$$

in words: x_I is a virtual mass-spring-damper being pushed around by the measured human force F_h — it is the trajectory a compliant admittance-causal robot would follow, computed by integrating this ODE forward rather than commanding a wrench directly. Regulating $p - x_I$ with an inner tracking loop, after canceling the robot's own dynamics, can expose a convenient double-integrator residual model. It is nevertheless *admittance-causal*: motion is the commanded quantity, force only drives the reference.

This paper does **not** use (8) as the physical controller — Section 4.B's impedance law is the actual commanded wrench, chosen specifically because it gives the clean physical-port power decomposition of Equation (7), which the energy-tank authorization of Section 5.2 depends on. Equation (8) is retained only as an **analytical intentional-response reference**: a stand-in for "how the interaction ought to feel," against which the impedance law's actual behavior is compared. This choice has a real cost: because the physical nominal is impedance-causal, the residual model that Section 5's MPC predicts against depends on the rendered stiffness, damping, and operational inertia (Section 4.D below), rather than being a gain-independent double integrator the way a literal admittance controller's residual would be. We do not claim gain-independent QP reuse.

4.D Error-based residual dynamics

The admittance reference (8) is driven only by F_h; written directly in task coordinates,

$$\ddot{x}_I = \Lambda^{-1}(F_h - D_I \dot{x}_I - K_I x_I). \tag{9}$$

in words: this is Equation (8), just expressed with the task-space inertia Λ already inverted through, so it can be integrated alongside the real robot state at every fast tick without ever being commanded to it. With $z = e - x_I$ — how far the impedance law's actual error is from where the admittance reference thinks it should be — the frozen local residual model used at each manager update is

$$\ddot{z} = \Lambda^{-1}(-K_I z - D_I \dot{z} + F_r + \hat{d}). \tag{10}$$

in words: z drifts according to the same impedance stiffness/damping, plus whatever residual wrench the MPC adds, plus the estimated disturbance $\hat{d}$ it is trying to reject. Unlike a literal admittance-reference construction, (10) explicitly depends on K_I, D_I, and $\Lambda(q)$ — the computational cost, flagged in 4.C, of retaining an impedance-causal physical nominal.

5. Predictive Proposal and Two Fast Authorizers

5.1 Residual MPC

At $T_m = 10$ ms, (10) is zero-order-hold discretized as

$$\xi_{k+1} = A_k \xi_k + B_k(F_{r,k} + \hat{d}_k), \qquad \xi = [z^\top, \dot{z}^\top]^\top. \tag{11}$$

The horizon-N proposal solves

$$\min_{F_{r,0:N-1}} \frac{1}{2} \sum_{i=1}^{N} \xi_i^\top Q \xi_i + \frac{1}{2} \sum_{i=0}^{N-1} F_{r,i}^\top R F_{r,i} \tag{12}$$

subject to a Cartesian wrench box and the current-model torque envelope

$$|\tau_I + J_v^\top F_{r,i}| \le \rho\tau_{\max}, \qquad \rho = 0.28. \tag{13}$$

The 28% envelope represents a deliberately derated continuous budget, chosen before the accepted run to keep the nominal torque feasible while preserving headroom pressure for the residual; it is not a manufacturer continuous-duty specification, and the absolute FR3 limits remain the MuJoCo safety backstop regardless. The benchmark rejects any trial in which τ_I itself is infeasible. Because (13) freezes J_v and τ_I, only the fast layer certifies the realized nonlinear sample.

5.2 Proposed finite-energy authorization

Let the tank obey, away from its upper cap,

$$\dot{E} = v^\top D_I v - F_r^\top v, \qquad E \ge E_{\min} > 0. \tag{14}$$

Proposition 1 (ideal composition). If (7) holds, the realized residual wrench is the same F_r used in (14), and an authorization mechanism maintains $E \ge E_{\min}$, then

$$\dot{H} + \dot{E} \le F^\top v. \tag{15}$$

Proof. Add (7) and (14). Residual power and nominal damping cancel. Energy discarded at the tank cap adds only nonnegative dissipation. □

At fast sample ℓ, the torque-feasible scale $\alpha_{\tau,\ell}$ is the largest value in $[0,1]$ satisfying

$$|\tau_{I,\ell} + J_{v,\ell}^\top(\alpha_{\tau,\ell} F_{r,k}^{\mathrm{raw}})| \le \rho\tau_{\max}. \tag{16}$$

Define $\bar{F}_{r,\ell} = \alpha_{\tau,\ell} F_{r,k}^{\mathrm{raw}}$. The energy scale is

$$\alpha_{E,\ell} = \begin{cases} 1, & \bar{F}_{r,\ell}^\top v_\ell \le 0, \\ \min\left(1, \dfrac{E_\ell - E_{\min} + h v_\ell^\top D_I v_\ell}{h \bar{F}_{r,\ell}^\top v_\ell}\right), & \text{otherwise.} \end{cases} \tag{17}$$

The applied wrench and ledger update are

$$F_{r,\ell} = \alpha_{E,\ell} \bar{F}_{r,\ell}, \tag{18}$$

$$E_{\ell+1} = \min\{E_{\max}, E_\ell + h(v_\ell^\top D_I v_\ell - F_{r,\ell}^\top v_\ell)\}. \tag{19}$$

Lemma 1 (fast-sample interface invariant). If $E_0 \ge E_{\min}$ and the nominal torque satisfies $|\tau_I| \le \rho\tau_{\max}$, then (16)–(19) ensure $E_\ell \ge E_{\min}$ and $|\tau_{I,\ell} + J_{v,\ell}^\top F_{r,\ell}| \le \rho\tau_{\max}$ at every fast sample.

Proof. Equation (16) constructs a feasible line segment from the feasible nominal torque. Further scaling by $\alpha_E \in [0,1]$ remains on that segment. For nonpositive residual power, (19) cannot reduce the ledger. For positive power, (17) limits withdrawal to the available energy above $E_{\min}$ plus the current damping contribution. □

5.3 External Hannaford–Ryu PO/PC baseline

The external baseline shares the same F_r^{raw}, nominal torque, torque scaling, estimator, and sample rate. Following the impedance-causal series PO/PC in [7], generalized from a scalar to the 3-D translational port, define

$$W_{\ell+1}^{\mathrm{pred}} = W_\ell - h\bar{F}_{r,\ell}^\top v_\ell. \tag{20}$$

If this prediction is negative, the controller adds the minimum isotropic damping

$$\gamma_\ell = \frac{-W_{\ell+1}^{\mathrm{pred}}}{h\|v_\ell\|^2}, \qquad F_{r,\ell} = \bar{F}_{r,\ell} - \gamma_\ell v_\ell; \tag{21}$$

otherwise $\gamma_\ell = 0$. A final torque-feasible scalar is applied and the observer is updated from the actual wrench. This is an equation-faithful translational generalization of [7], with two disclosed adaptations: the published scalar port is extended by $\|v\|^2$, and all output is passed through the same FR3 torque interface. It has zero initial energy reference, whereas the proposed tank permits finite stored energy and harvesting of nominal damping. This is the intended scientific comparison.

6. Benchmark Design

6.1 Plant, controllers, and signals

The plant is the torque-controlled Franka FR3 model from MuJoCo Menagerie, integrated at 1 kHz. Built-in position actuators are disabled; the benchmark applies joint torque directly. The four controllers are:

- **B1 Passive impedance:** equations (3)–(4), $F_r = 0$;
- **B2 Unguarded MPC:** equations (11)–(13), with fast torque scaling but no residual-energy restriction;
- **B3 Hannaford–Ryu PO/PC:** the same raw MPC plus (20)–(21);
- **B4 Two-rate tank:** the same raw MPC plus (16)–(19).

All controllers use $K_I = 180$ N/m, $D_I = 28$ N s/m, a 100 Hz manager, a 1 kHz torque loop, $N = 20$ (0.20 s), $E_0 = 0.08$ J, $E_{\min} = 0.02$ J, and $E_{\max} = 0.30$ J. The intentional force contains an 8 N push along x and a -5 N push along z. Rejectable force contains three sinusoids and a 12 N, 7 ms pulse beginning at 1.507 s, between manager ticks. A unilateral passive wall starts 35 mm from the nominal pose.

6.2 Matched randomized protocol

Twenty matched trials randomize:

- passive-wall stiffness: 500–1500 N/m;
- passive-wall damping: 8–20 N s/m;
- rejectable-force scale: 0.8–1.2; and
- the three sinusoidal phases.

Every controller receives identical realization parameters for each seed. Main metrics are 3-D residual-position RMS/peak, minimum energy ledger, PO energy, authorization activity, peak fraction of the derated joint-torque envelope, nominal infeasibility, and QP failures. Paired statistics are computed over seeds. Solver timing covers only `OSQP.solve`, not model construction, sensing, or torque communication.

6.3 Force-separation leakage test

The main benchmark assumes exact force labels. The leakage test directly relaxes that assumption:

$$\hat{d} = d + F_e + \lambda F_h + n_F, \qquad \lambda \in \{0, 0.1, 0.25, 0.5\}. \tag{22}$$

To isolate leakage from disturbance rejection, this test disables d and the wall, retains 0.05 N estimator noise, and runs five matched seeds per level. It reports RMS error along the intentional-force axis and the ratio between realized and reference mean displacement during the sustained push.

A second, satellite sweep at a fixed mid-range leakage ($\lambda = 0.25$) adds three sensing imperfections the main sweep above holds at their simplest setting: a one-manager-tick (10 ms) estimator delay, AR(1) colored noise ($\phi = 0.9$, same stationary standard deviation as the white-noise baseline so only temporal correlation is varied), and a constant 5 mm/s velocity-estimate bias along the intentional-force axis, applied only to the state fed to the residual MPC (the torque loop itself still uses the true measured velocity). Each is toggled individually and then combined, five matched seeds per condition.

7. Results

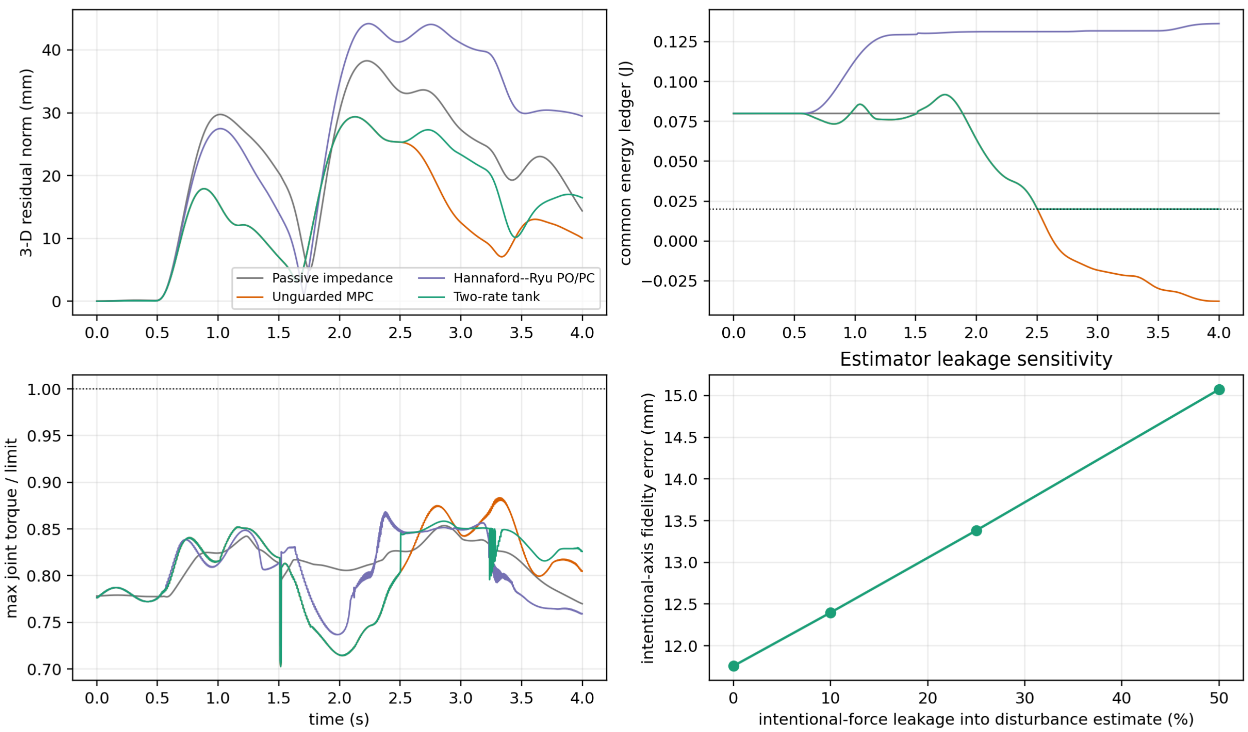

Figure 1: Torque-controlled FR3 response, energy audit, torque utilization, and force-separation leakage.

7.1 Matched FR3 benchmark

Table 1. Twenty matched FR3 trials, mean ± sample standard deviation. Torque ratio is relative to the derated 28% continuous envelope.

Controller	residual RMS (mm)	residual peak (mm)	minimum ledger/PO (J)	authorization active	peak torque ratio
Passive impedance	25.42 ± 0.76	37.07 ± 1.44	0.0800 ± 0.0000	0%	0.853 ± 0.002
Unguarded MPC	**16.23 ± 0.92**	**28.42 ± 1.68**	-0.0424 ± 0.0137	0%	0.875 ± 0.005
Hannaford–Ryu PO/PC	30.00 ± 0.92	43.62 ± 1.79	-1.3×10^{-19} PO	83.36 ± 1.79%	0.867 ± 0.006

Controller	residual RMS (mm)	residual peak (mm)	minimum ledger/PO (J)	authorization active	peak torque ratio
Two-rate tank	19.39 ± 0.61	31.21 ± 1.80	**0.0200 ± 0.0000**	29.79 ± 4.46%	0.865 ± 0.010

The unguarded controller's counterfactual common ledger crosses the 0.02 J floor in 20/20 trials. This does not mean a physical tank becomes negative; it measures energy the controller would withdraw without authorization. The PO/PC observer remains nonnegative to numerical precision in every trial, and the proposed tank never crosses its floor.

The proposed controller reduces residual RMS by 23.7% relative to passive impedance (paired difference -6.024 mm, 95% CI [-6.500, -5.549], $p = 1.80 \times 10^{-16}$). It reduces RMS by 35.4% relative to strict PO/PC (difference -10.605 mm, 95% CI [-10.985, -10.225], $p = 6.61 \times 10^{-23}$). Unguarded MPC remains 19.5% better than the proposed method (proposed-minus-unguarded +3.166 mm, 95% CI [2.637, 3.695], $p = 1.25 \times 10^{-10}$). Thus, finite energy storage recovers much of the performance removed by zero-reference PO/PC but does not eliminate the energetic cost of passivity-oriented authorization.

Every accepted trial has zero nominal-torque infeasibility and zero QP failure. The largest torque ratio is below 0.889 for the proposed controller. Mean solver-core time is 0.159 ms; the mean per-run 95th percentile is 0.179 ms and the largest recorded solve is 0.399 ms. The 100 Hz deadline is therefore met by the solver core, but this is not an end-to-end timing certificate.

7.2 Force-separation leakage

Table 2. Intentional-force leakage, five matched noise seeds per level.

leakage λ	intentional-axis error RMS (mm)	realized/reference response ratio	minimum tank (J)
0	11.759 ± 0.007	0.741	0.020
0.10	12.399 ± 0.007	0.724	0.020
0.25	13.383 ± 0.007	0.698	0.020
0.50	15.073 ± 0.006	0.655	0.020

At 50% leakage, fidelity error is 28.2% higher than at zero leakage and the mean response ratio falls by 8.6 percentage points. The tank and torque invariants still hold; the failure is semantic rather than numerical. The controller safely does the wrong thing because part of the intentional human input is mislabeled as a disturbance. This result makes force decomposition a first-order interface requirement rather than a footnote.

Table 3. Sensing-realism sweep at fixed $\lambda = 0.25$ leakage, five matched seeds per condition.

Condition	intentional-axis error RMS (mm)	response ratio	minimum tank (J)
Baseline ($\lambda = 0.25$, no added realism)	13.384 ± 0.008	0.698	0.020
Delay only (10 ms)	13.368 ± 0.008	0.699	0.020
Colored noise only	13.374 ± 0.039	0.699	0.020
Velocity bias only (5 mm/s)	13.730 ± 0.008	0.690	0.020
All combined	13.704 ± 0.039	0.690	0.020

The baseline row is this sweep's own $\lambda = 0.25$ draw (different seeds from, but statistically consistent with, Table 2's 13.383 mm). None of the three individually degrades fidelity error by more than 2.6% relative to baseline, and the tank floor and torque envelope hold in every condition – the safety certificates are not sensitive to these particular sensing imperfections at these levels. The one-tick estimator delay and the colored-noise correlation structure leave the mean essentially unchanged; colored noise instead widens the across-seed standard deviation roughly fivefold (0.008 to 0.039 mm), because temporally correlated noise resists the horizon's implicit averaging. The velocity-estimate bias is the dominant of the three, degrading fidelity error by 2.6% and the response ratio by 0.9 percentage points, because it corrupts the state the residual MPC itself feeds back on, not merely the disturbance estimate. The combined condition tracks the velocity-bias-only condition closely rather than summing the three effects, mirroring the leakage sweep's own message: safety (tank, torque) survives these corruptions, but task fidelity degrades in proportion to which channel is corrupted, and no causal claim beyond the tabulated numbers is made for why the combined condition does not exceed the velocity-bias-only one.

7.3 Manager-rate sensitivity: no universal winner

Sections 6-7 fix the manager at 100 Hz. To check whether that choice was load-bearing, the main matched benchmark, the leakage sweep, and the sensing-realism sweep were rerun at 20 Hz and 50 Hz as well, holding the horizon duration fixed at 0.20 s throughout – only the re-check interval changes (50, 20, or 10 ms), not the manager's lookahead.

Table 4. Manager-rate sweep, main matched benchmark, two-rate controller only (20 seeds).

Manager rate	RMS (mm)	Peak (mm)	Authorization active	vs. unguarded MPC	Solve max / period
20 Hz	18.96 ± 0.53	30.71	17.2%	+5.1%	0.140 ms / 50 ms
50 Hz	19.50 ± 0.70	31.58	27.4%	+16.3%	0.169 ms / 20 ms
100 Hz	19.39 ± 0.61	31.21	29.8%	+19.5%	0.285 ms / 10 ms

In this oscillatory wall-contact-plus-disturbance scenario, 20 Hz gives the lowest RMS, the least-frequent intervention, and (in this run) the smallest computational burden relative to its own deadline; solve-time maxima are wall-clock and have been observed to vary run-to-run by 2-3× at a fixed rate, so `rate_sweep.py`'s own output is the source of record rather than this table's specific figures. A representative trial's diagnostics explain the ranking, not just describe it: at 20 Hz the applied correction has a total variation of 12.9 N/s and 0.75 activation events per second, versus 34.6 N/s and 4.25 events per second at 100 Hz – almost triple the chatter. More tellingly, at 20 Hz tracking is *better* while the tank is active than while idle (15.2 versus 18.3 mm RMS); at 100 Hz this reverses (20.6 versus 16.1 mm). A slower manager commits to fewer, larger, apparently more coherent corrections in this scenario; a faster one re-solves before the previous correction has had time to act, producing more frequent, less decisive adjustments – a genuine chatter mechanism, not a fluke of one metric.

Table 5. Manager-rate sweep, force-separation leakage (5 seeds per level).

Manager rate	$\lambda = 0$ (mm)	$\lambda = 0.5$ (mm)	% degradation
20 Hz	13.01 ± 0.03	17.21 ± 0.03	32.3%
50 Hz	12.06 ± 0.01	15.60 ± 0.01	29.3%
100 Hz	11.76 ± 0.01	15.07 ± 0.01	28.2%

Under this different scenario – a sustained intentional push with no wall or oscillatory disturbance – the ranking **reverses**: 100 Hz gives the lowest error at every leakage level and 20 Hz the highest. Tracking a step-like sustained push is a settling-time problem rather than a disturbance-rejection problem, so a slower supervisory loop is plausibly sluggish to lock onto the correct steady correction – the opposite failure mode from the oscillatory-disturbance case above. The sensing-realism sweep shows the same qualitative pattern (velocity bias dominant, colored noise widens variance, delay negligible) at all three rates, only shifted to each rate's own baseline; that shift is consistent with, not independent evidence for, the ranking in this table.

No manager rate dominates in both regimes tested, and neither compromises safety. Zero tank violations, zero QP failures, and zero nominal infeasibilities hold at every rate in every scenario reported in this paper. The 100 Hz design point used throughout Sections 6-7.2 is the better choice for the sustained-push/leakage regime this paper otherwise emphasizes, but Table 4 shows it is not the best choice for the oscillatory wall-contact scenario, where a slower manager measurably reduces both intervention frequency and tracking error. Matching manager rate to the anticipated disturbance's own time scale – rather than assuming faster is always safer – is a real design question this benchmark exposes rather than resolves.

7.4 Anticipatory disturbance forecasting: the harmonic model fails on non-harmonic content

The main benchmark's disturbance term (Section 6.1) contains three sinusoids of known, fixed frequency (0.9, 1.4, 1.9 Hz). This raises a natural question: if those frequencies are known in advance – a disclosed, benchmark-specific modeling choice, not a general online frequency-identification claim – can the MPC do better than holding its disturbance estimate constant across the horizon (the frozen zero-order hold used everywhere else in this paper)? An **anticipatory** variant tracks each axis with a small per-axis Kalman filter over the two-dimensional sine/cosine state at the known frequency and forecasts that fitted model forward through the full 0.20 s horizon, replacing the frozen hold on $\hat{d}_k$ in equation (11) with a per-step forecast; everything else – the same tank, the same torque authorization, the same measurement – is unchanged.

Every test condition below differs in how much of the measured disturbance is genuinely representable by a two-parameter (amplitude/phase) sinusoid at a known frequency, and how much is not: a brief non-periodic pulse, or a contact force with no fixed frequency at all. Four wall-disabled, twenty-seed conditions isolate this axis from contact: a **pure-harmonic ablation** (the three sinusoids only, the 12 N pulse suppressed via pulse_scale = 0 – zero non-harmonic content, the cleanest case the forecaster could ask for), **periodic-only** (the scenario used throughout Sections 6-7, sinusoids plus the pulse both active – a small amount of non-harmonic content), and **pulse-only** (sinusoids silenced, the pulse alone active – entirely non-harmonic content, the negative-control extreme). A fifth, satellite check reruns the full matched benchmark (wall active, everything on – substantial non-harmonic content from contact) to see whether any advantage survives once contact force dominates the disturbance estimate.

Table 6. Anticipatory (harmonic forecast) vs. two-rate (frozen hold), paired by seed, ordered by increasing non-harmonic content.

Condition	two-rate RMS (mm)	anticipatory RMS (mm)	paired difference (mm)	95% CI	p
Pure-harmonic ablation (pulse removed)	20.234 ± 1.684	20.239 ± 1.625	+0.005	[-0.087, 0.098]	9.04×10^{-1}
Periodic-only (as tested, Sections 6-7; small pulse present)	**20.409 ± 1.679**	20.737 ± 1.628	+0.328	[0.220, 0.436]	4.24×10^{-6}
Pulse-only† (entirely non-harmonic)	**21.212 ± 0.026**	21.583 ± 0.010	+0.371	[0.359, 0.383]	8.72×10^{-24}

Condition	two-rate RMS (mm)	anticipatory RMS (mm)	paired difference (mm)	95% CI	p
Full matched (satellite; substantial contact content)	19.133 ± 0.634	**18.773 ± 0.457**	-0.360	[-0.541, -0.179]	5.23×10^{-4}

†With the sinusoids silenced (disturbance_scale = 0), the per-seed random phases no longer reach the trial; without another source of per-seed variation, all 20 seeds would replay an identical trajectory and any paired statistic here would be spurious. This scenario adds the same 0.05 N estimator noise used by the leakage and sensing-realism sweeps (Section 6.3) as its sole source of seed-to-seed variation, giving the 20 seeds genuine independence.

The ablation row is the key control: with the pulse fully removed, leaving content the harmonic model can represent exactly, anticipatory and frozen-hold tracking are statistically indistinguishable ($p = 0.90$). Add back the same 12 N, 7 ms pulse that is present throughout the rest of this paper's periodic-only scenario, and a small but highly significant degradation appears (+1.6%, $p = 4.24 \times 10^{-6}$) – a single brief non-harmonic event is enough to measurably hurt the forecaster, even though it occupies 7 ms of a 4 s trial. Remove the harmonic content entirely and keep only that same pulse (pulse-only), and the degradation is of comparable size (+1.75%) but now overwhelmingly significant ($p = 8.72 \times 10^{-24}$), because every one of the 20 seeds sees the identical, unmodeled transient. The pattern is monotonic in non-harmonic content, not in disturbance complexity generally: the harmonic filter does what it is built to do when the signal is genuinely harmonic, and is actively misled whenever it is not.

The ablation's near-zero point estimate does not mean the pure-harmonic case is *exactly* neutral, only that this benchmark's 20 seeds cannot distinguish it from neutral. A real, small mechanism is identifiable underneath it: fed the same noiseless sinusoid the simulator generates, the per-axis Kalman filter's forecast error settles to the order of 10^{-6} mm within 0.5 s (so the residual effect is not a prediction-accuracy failure), and directly inspecting the QP at a disturbance peak shows that, given an accurate forecast that the disturbance is about to swing back toward zero, the optimizer commits *less* immediate corrective wrench than the frozen-hold controller, which incorrectly assumes the disturbance persists at its current value for the full horizon and therefore always corrects harder right now. Because the horizon's stage cost (equation (12)) weights every step $\xi_i^\top Q \xi_i$ equally, with no discount or terminal emphasis on the near term, the MPC has no structural reason to prefer the immediate error the RMS metric actually measures over the horizon-averaged cost it is actually minimizing. This foresight/reactivity effect is real but second-order here: a full blend-parameter sweep on the as-tested (pulse-present) periodic-only scenario degrades RMS *monotonically* – 20.41, 20.48, 20.56, 20.65, 20.74 mm at blend $0, 0.25, 0.5, 0.75, 1$ between the frozen hold and the full forecast – but as the ablation shows, most of that headline gap is the pulse, not this mechanism alone.

A natural objection is that 0.20 s ($N = 20$) simply is not enough lookahead – perhaps a longer horizon lets the controller collect more benefit than cost. An eight-seed horizon sweep (`horizon_sweep.py`, $N \in \{10, 20, 40, 60, 80, 110\}$, i.e. 0.10-1.10 s – past a full period of the slowest, 0.9 Hz sinusoid) rules this out, and separates the two mechanisms cleanly:

Table 7. Horizon-length sweep, periodic-only, eight seeds (mean RMS; not the paired 95% CI/p that Table 6's twenty seeds support).

Horizon N (lookahead)	two-rate RMS (mm)	anticipatory RMS (mm)	relative change	pure-harmonic ablation, relative change
10 (0.10 s)	23.877	23.941	+0.3%	–

Horizon N (lookahead)	two-rate RMS (mm)	anticipatory RMS (mm)	relative change	pure-harmonic ablation, relative change
20 (0.20 s, paper default)	20.739	21.156	+2.0%	+0.4%
40 (0.40 s)	20.091	20.868	+3.9%	–
60 (0.60 s)	20.081	20.879	+4.0%	–
80 (0.80 s)	20.085	20.847	+3.8%	–
110 (1.10 s)	20.094	20.840	+3.7%	+1.0%

The as-tested (pulse-present) gap widens sharply with lookahead – from 0.3% at 0.10 s to 3.9% by 0.40 s – then plateaus near 3.7-4.0% out to 1.10 s. Repeating the two horizon endpoints with the pulse removed (last column, eight seeds each) shows the pure-harmonic effect also grows with lookahead, but far more slowly and from a much smaller base (+0.4% to +1.0%, versus +2.0% to +3.7% with the pulse present) – consistent with two compounding effects, not one: a small, genuine foresight/reactivity cost that scales mildly with horizon even for a perfectly-fit sinusoid, and a much larger cost from a longer horizon leaning harder on the still-recovering, pulse-corrupted forecast. A longer horizon is not the fix for either.

The full-matched satellite check (Table 6) flips the sign at the paper's own $N = 20$: there the anticipatory controller is 1.9% *better*. That advantage does not survive the same horizon sweep repeated under full-matched conditions (wall active, everything on):

Table 8. Horizon-length sweep, full matched, eight seeds (mean RMS only).

Horizon N (lookahead)	two-rate RMS (mm)	anticipatory RMS (mm)	relative change
10 (0.10 s)	22.528	22.416	-0.5%
20 (0.20 s, paper default)	19.184	18.834	-1.8%
40 (0.40 s)	18.226	19.952	+9.5%
60 (0.60 s)	17.890	20.144	+12.6%
80 (0.80 s)	18.181	19.746	+8.6%
110 (1.10 s)	18.357	19.749	+7.6%

At $N = 10$ and $N = 20$, anticipatory is modestly better (-0.5%, -1.8% – the latter consistent with Table 6's -1.9% at $N = 20$ within this smaller sample's noise); at $N = 40$ it reverses to a 9.5% disadvantage and stays substantially worse (7.6-12.6%) through $N = 110$. **The full-matched advantage reported in Table 6 is therefore a narrow, horizon-specific artifact of the paper's own $N = 20$ design point, not a robust benefit of forecasting under contact.** The same non-harmonic-content mechanism explains why, at a larger scale: the wall-contact force is exactly the kind of content a two-parameter sinusoid cannot represent, and an instrumented, single-seed trace of the predictor's own fitted state shows the damage is severe and persistent – filter error against the true periodic component rises from 0.037 mm (periodic-only, uncontaminated) to 4.56 mm during contact, and has not recovered even 0.30 s after contact ends (0.929 mm, still roughly 30× the uncontaminated case). A longer horizon leans harder on exactly this contaminated forecast, so the same mechanism that costs a few percent against a 7 ms pulse costs closer to 10% against a sustained contact force. The pulse-only row (Table 6) is a genuine negative control – the harmonic model has no periodic content to lock onto – and it, too, is significantly worse for the anticipatory controller, consistent with this finding rather than merely directionally similar to it.

The accuracy question aside, the anticipatory forecaster is not free to build. It adds a disclosed but non-trivial modeling assumption (the disturbance frequencies must be known in advance), three new tunable Kalman-filter hyperparameters (process noise, measurement noise, initial variance) that the frozen hold needs none of, and 131 new lines of code (`harmonic_disturbance_predictor.py` plus its test) atop a 52-line integration diff into the benchmark's controller dispatch. None of that cost is computational: the predictor's per-tick update-and-forecast call measures 35.5 μs against a full MPC solve's 2143.5 μs (about 1.7% overhead), and end-to-end per-trial wall-clock time is statistically indistinguishable between the two controllers (2.51 s vs. 2.40 s across five representative trials, well within run-to-run noise) – so nothing here argues for the frozen hold on a compute budget. The argument is that the added assumption and machinery are paid for in every condition this paper tests – neutral in the one case built to be maximally favorable to it (pure-harmonic ablation), and a measurable loss everywhere real disturbances are less than perfectly harmonic – while the frozen hold needs no frequency assumption, no additional filter, and no per-condition retuning to reach the same or better tracking.

7.5 Interpretation

The benchmark supports three claims. First, finite-horizon residual correction can improve the realized intentional impedance response under disturbances. Second, strict zero-reference PO/PC is substantially more conservative than a finite tank that can spend initial and dissipated energy. Third, the fast layer can preserve its explicit tank and torque interfaces even when the force classifier is wrong, so those certificates must not be confused with task or intent correctness. A fourth, unplanned finding (Section 7.3) is that no single manager rate is best across scenario types – the right rate depends on whether the disturbance is oscillatory or a sustained step, and the paper's own headline rate is a choice for one of those regimes, not a universal optimum. A fifth, also unplanned finding (Section 7.4) is that a harmonic disturbance forecaster is neutral only when the disturbance is genuinely harmonic, and measurably worse once it is not: a controlled ablation isolates a small, real foresight/reactivity cost even for a perfectly-fit sinusoid, but the dominant driver in every other condition tested is non-harmonic content – a brief pulse, or contact – that a two-parameter harmonic model cannot represent, and that a longer horizon leans on more, not less, heavily. The one condition where forecasting appeared to help (wall contact present) does not survive that same horizon check either, and the added modeling assumption and filter machinery cost real engineering complexity for no compute saving, so no configuration tested in this paper supports disturbance forecasting as a robust improvement over the frozen hold.

The comparison does not establish superiority over passive model-predictive impedance [2], predictive variable-impedance methods [3]–[5], or predictive admittance [6]. Those controllers optimize different quantities and several have physical experiments. The comparison in Section 3 is architectural, not a numerical ranking.

8. Scope and Limitations

Validation is nonlinear 7-DoF rigid-body simulation, not hardware, and contact is a virtual unilateral passive wall rather than measured material interaction; there are no human participants, contact-force safety thresholds, or claims of clinical/industrial readiness. Proposition 1 is an ideal continuous-time composition statement, and Lemma 1 certifies the implemented energy and torque interfaces at fast samples, not a global sampled-data passivity theorem for the entire robot, orientation task, null-space controller, estimator, and environment; similarly, the MPC horizon freezes the current Jacobian, operational inertia, and nominal torque, so the fast projection enforces realized input feasibility but recursive MPC feasibility and state constraints are not proved. The Hannaford–Ryu baseline is equation-faithful at the translational port but necessarily generalized from a scalar haptic interface and passed through the same FR3 torque projection, so it is not a reproduction on the original Excalibur device. The main force labels are exact; the leakage study quantifies one failure mode, and Section 7.2 additionally tests estimator delay, colored noise, and a constant velocity-estimate bias, individually and combined, at one representative leakage level – but not as a learned human-intent estimator, and not swept jointly across every leakage level. The anticipatory forecaster (Section 7.4) assumes the disturbance frequencies are known exactly, matched to this benchmark's own simulator-generated frequencies – a disclosed modeling choice, not a general online frequency-identification method. Its one favorable result (the full-matched scenario at the paper's own 0.20 s horizon) reverses under a longer-horizon check, so it should not be read as evidence that forecasting helps under contact; the horizon sweeps behind that finding use a smaller, eight-seed exploratory sample rather than the paired twenty-seed statistic Table 6 otherwise reports, and the single-seed filter-error trace and per-tick compute-overhead measurement cited alongside are exploratory diagnostics, not committed, reproducible sources.

9. Conclusion

This paper retains an impedance-causal physical nominal and assigns MPC only an additive residual-wrench port. The structure sacrifices gain-independent residual matrices but exposes the exact power that must be authorized. A 100 Hz predictive proposal is therefore paired with a 1 kHz projection that accounts for measured residual power and complete joint-torque headroom.

The torque-controlled FR3 study adds two pieces of evidence absent from the earlier low-order verification: an external, equation-faithful time-domain passivity baseline and a direct intent/disturbance leakage test. The finite tank occupies a useful middle ground between unguarded MPC and strict zero-reference PO/PC, while the leakage results show that energy correctness cannot substitute for correct interpretation of human input. A manager-rate sweep (Section 7.3) further shows that the 100 Hz design point used throughout is a choice suited to sustained-push tracking, not a universally optimal rate: an oscillatory-disturbance scenario is better served by a slower, less chatter-prone manager, and the two regimes disagree on which rate wins. An anticipatory-forecasting variant (Section 7.4) shows that horizon foresight is not a free improvement either: a controlled ablation finds it statistically neutral for genuinely harmonic content, but every departure from that – a brief pulse, contact – costs measurable tracking accuracy that a longer horizon widens rather than closes, at real added modeling and code complexity for no compute saving. The one scenario where forecasting looked beneficial (wall contact present) turned out to be specific to the paper's own 0.20 s horizon and reversed at longer horizons, so this paper finds no configuration in which disturbance forecasting robustly helps with a single, fixed predictor and horizon. Table 8's own best horizon for full-matched ($N = 10$) differs from the horizon that best serves periodic-only, suggesting a regime-adaptive scheme – detecting whether the current disturbance content is harmonic or not and switching predictor and horizon accordingly – as a candidate follow-up; nothing in this paper's results shows such a scheme would beat the frozen hold rather than merely match it, so this is future work, not a claim. The next decisive validation is a torque-controlled FR3/Panda contact experiment with measured wrench, velocity noise, latency, and complete end-to-end timing.

Reproducibility

From this directory:

```
cd simulation

MPLCONFIGDIR=/tmp/phri_impedance_fr3_mpl \
XDG_CACHE_HOME=/tmp/phri_impedance_fr3_mpl \
python3 verify_fr3_two_rate_benchmark.py --seeds 20 --leakage-seeds 5 --realism-seeds 5

python3 rate_sweep.py --leakage-seeds 5 --realism-seeds 5

python3 anticipatory_disturbance_study.py --seeds 20

python3 horizon_sweep.py --seeds 8

python3 -m pytest -q \
  test_fr3_two_rate_benchmark.py \
  test_two_rate_passive_residual.py \
  test_residual_mpc.py \
  test_rate_sweep.py \
  test_harmonic_disturbance_predictor.py \
  test_horizon_sweep.py
```

All simulation, verification, and test scripts live in `simulation/`. The primary script writes `simulation/fr3_two_rate_resul` and `simulation/fr3_two_rate_results.png`. `rate_sweep.py` produces `simulation/rate_sweep_results.json`, the source for Table 4/5 and the manager-rate diagnostics of Section 7.3. `anticipatory_disturbance_study.py` produces `simulation/anticipatory_disturbance_results.json`, the source for all four rows of Table 6 (including the pure-harmonic ablation) – the blend sweep and QP-peak inspection discussed alongside it were exploratory and are not part of this repro command. `horizon_sweep.py` produces `simulation/horizon_sweep_results.json`, the source for Tables 7-8, including Table 7's pure-harmonic ablation column; the single-seed predictor-error trace and the per-tick compute-overhead measurement cited in Section 7.4 are further exploratory diagnostics not captured by any committed script. The earlier 1-DoF benchmark remains available as a secondary algebra and inter-update audit. `state_of_art_search_log.md` records the literature-search scope and novelty decision.

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